Notebook 6 — Chapter 6.4

Optimal Capital Growth & Position Sizing

21 worked examples — Kelly criterion, fractional Kelly, volatility drag, gambler's ruin and drawdown risk — calibrated for Stage 2 speed and rigor

Core Toolkit (read this first)

Kelly: bet the fraction f of wealth that maximises the long-run growth rate $G(f) = \mathbb{E}[\ln(1+fX)]$, because $\ln W_n = \sum \ln(1+fX_i)$ and the law of large numbers gives $\frac{1}{n}\ln W_n \to G$.
Win $+b$ with prob. p, lose -1 with prob. $q = 1-p$:

$$G(f) = p\ln(1+bf) + q\ln(1-f), \qquad f^\star = \frac{bp-q}{b} = p - \frac{q}{b} = \frac{\text{edge}}{\text{odds}}$$

Return $+a$ w.p. p, $-d$ w.p. q: $f^\star = \frac{p}{d} - \frac{q}{a}$. Gaussian asset (excess mean μ, vol σ): $g(f) = f\mu - \frac{1}{2}f^2\sigma^2$, $f^\star = \frac{\mu}{\sigma^2}$, $g^\star = \frac{\mu^2}{2\sigma^2}$.
Fractional Kelly $cf^\star$: $G(cf^\star) \approx G^\star\, c(2-c)$; growth ≈ 0 at $2f^\star$.
Ruin ($r = q/p$, unit bets): hit N before 0 from k: $\frac{1-r^k}{1-r^N}$; ruin ever (if $p > q$): r^k; fair game: $\frac{k}{N}$ and $\mathbb{E}[T] = k(N-k)$. **Drawdown:** $\mathbb{P}(\text{ever} \le \alpha W_0) \approx \alpha^{2/c-1}$ at c·Kelly.

A. The Kelly Criterion

Example 1 — Why Maximise $\ln W$? The Even-Money Bet

You repeatedly bet a fraction f of your bankroll on an even-money bet ($b = 1$) won with probability $p = 0.6$. Find the optimal f and the growth rate $G^\star$.
Solution. After a win wealth is multiplied by $1+f$, after a loss by $1-f$:

$$G(f) = 0.6\ln(1+f) + 0.4\ln(1-f)$$

$$G'(f) = \frac{0.6}{1+f} - \frac{0.4}{1-f} = 0 \;\Rightarrow\; 0.6(1-f) = 0.4(1+f) \;\Rightarrow\; \boxed{f^\star = 0.2}$$

$$G^\star = 0.6\ln 1.2 + 0.4\ln 0.8 = 0.6(0.1823) + 0.4(-0.2231) = \boxed{0.02014}$$

Closed form for even odds: $G^\star = \ln 2 + p\ln p + q\ln q$.
Why the log? Repeated bets *multiply* wealth, and ln turns a product into a sum, so the typical path grows like e^{nG}. Maximising $\mathbb{E}[W]$ instead would say "bet everything" ($f = 1$), which is ruined at the first loss.
⋆ *Analogy:* compound growth is a *relay race*: the team's finishing time is set by the *sum of the logs of the legs*, not the average speed of the fastest leg.

Example 2 — Deriving $f^\star = \frac{bp-q}{b}$

Derive the Kelly fraction for a bet that pays b-to-1 with win probability p, then evaluate $b = 2$, $p = 0.45$.
Solution. $G(f) = p\ln(1+bf) + q\ln(1-f)$ is concave ($G'' < 0$), so set $G' = 0$:

$$\frac{pb}{1+bf} = \frac{q}{1-f} \;\Rightarrow\; pb(1-f) = q(1+bf)$$

$$pb - q = f\,(pb + qb) = fb \quad (\text{since } p+q=1) \;\Rightarrow\; f^\star = \frac{bp-q}{b}$$

For $b = 2$, $p = 0.45$: edge $bp - q = 0.9 - 0.55 = 0.35$:

$$f^\star = \frac{0.35}{2} = \boxed{0.175}$$

$$G^\star = 0.45\ln 1.35 + 0.55\ln 0.825 = 0.13505 - 0.10580 = \boxed{0.0292}$$

▶ *Speed trick:* $f^\star = p - \frac{q}{b}$ is faster: $0.45 - \frac{0.55}{2} = 0.175$.

Example 3 — Edge Over Odds, and When Not to Bet

Compute the Kelly fraction for (*a*) $p = 0.4$, $b = 1.2$; (*b*) $p = 0.5$, $b = 3$; (*c*) $p = 0.3$, $b = 4$.
Solution. Use $f^\star = \frac{bp-q}{b} = \frac{\text{edge}}{\text{odds}}$.
(*a*) Edge $= 1.2(0.4) - 0.6 = -0.12 < 0$:

$$f^\star = \boxed{0} \quad \text{(no bet: negative expected value)}$$

(*b*) Edge $= 3(0.5) - 0.5 = 1$, so $f^\star = \frac{1}{3}$.
(*c*) Edge $= 4(0.3) - 0.7 = 0.5$, so $f^\star = \frac{0.5}{4} = \boxed{0.125}$: a bet you lose 70% of the time is still worth 12.5% of your bankroll.
⋆ *Analogy:* the stake follows the *edge per unit of odds*, like fuel scaled to how much ground the engine can cover: a long-shot with a big prize needs a small stake even if the edge is large.

Example 4 — The Growth Curve and the Cliff on the Right

For Example 1 ($p = 0.6$, even odds) compute $G(f)$ at $f = 0.05, 0.1, 0.2, 0.3, 0.4, 0.5$.
Solution. Evaluate $G(f) = 0.6\ln(1+f) + 0.4\ln(1-f)$:

f	.05	.10	.20	.30	.40	.50
G	.00876	.01504	.02014	.01475	−.00245	−.03398

Half-Kelly ($f = 0.1$) earns $\frac{0.01504}{0.02014} = 74.7\%$ of the maximum growth. At $2f^\star = 0.4$ growth is slightly negative; the exact zero-growth fraction is $f \approx 0.389$. At $f = 0.5$ the bettor *loses* 3.4% per bet despite a 60% win rate.
▶ *Speed trick:* the curve is flat near the top, falls gently to the left and sharply to the right. When unsure, bet *less*.
⋆ *Analogy:* cruising speed on a mountain road: slow driving only costs time, but a little too fast takes you off the cliff.

Example 5 — Mean Wealth Versus Typical Wealth

With $f = 0.2$ on the $p = 0.6$ even-money bet, find the *expected* wealth and the *median* wealth after 100 bets, and $\mathbb{P}(W_{100} < 1)$.
Solution. Expected: per bet $\mathbb{E}[1+fX] = 1 + 0.2(0.6-0.4) = 1.04$ and bets are independent:

$$\mathbb{E}[W_{100}] = 1.04^{100} = \boxed{50.5}$$

Typical: $\ln W_{100} \approx 100\,G^\star$, so the median is

$$e^{100(0.02014)} = e^{2.014} = \boxed{7.49}$$

(simulation of 2×10^5 paths: median 7.49, mean 51). Losing money requires at most 55 wins in 100 bets (since $55\ln 1.2 + 45\ln 0.8 < 0$):

$$\mathbb{P}(W_{100} < 1) = \mathbb{P}(\text{Bin}(100, 0.6) \le 55) = \boxed{17.9\%}$$

⋆ *Analogy:* the average lottery-ticket holder is rich because of one lucky winner; the *typical* holder is poor. Expected wealth is carried by rare extreme paths.

Example 6 — Two-Outcome Returns: +50% or −30%

An investment returns +50% or −30% with equal probability. What fraction of capital should be invested, and what is the growth rate?
Solution. With fraction f invested, wealth is multiplied by $1+0.5f$ or $1-0.3f$:

$$G(f) = \tfrac{1}{2}\ln(1+0.5f) + \tfrac{1}{2}\ln(1-0.3f)$$

$$G' = 0: \quad \frac{0.5(0.5)}{1+0.5f} = \frac{0.5(0.3)}{1-0.3f} \;\Rightarrow\; 0.25(1-0.3f) = 0.15(1+0.5f)$$

$$0.1 = 0.15f \;\Rightarrow\; f^\star = \boxed{\tfrac{2}{3}}$$

Check with $f^\star = \frac{p}{d} - \frac{q}{a} = \frac{0.5}{0.3} - \frac{0.5}{0.5} = \frac{5}{3} - 1 = \frac{2}{3}$. Growth:

$$G\left(\tfrac{2}{3}\right) = \tfrac{1}{2}\ln\tfrac{4}{3} + \tfrac{1}{2}\ln 0.8 = \boxed{0.0323}$$

Equivalently, in "risk" terms the bet is a $b = \frac{0.5}{0.3} = \frac{5}{3}$ payoff and the risked fraction is $fd = 0.2 = p - \frac{q}{b} = 0.5 - 0.3$.
▶ *Speed trick:* $f^\star = \frac{p}{d} - \frac{q}{a}$ is Kelly with the loss d as the unit of risk.

Example 7 — Kelly for a Gaussian Asset: $f^\star = \mu/\sigma^2$

An asset has excess return $\mu = 8\%$ per year and volatility $\sigma = 20\%$. Find the Kelly leverage, the growth rate, and the growth at half-Kelly and at twice Kelly.
Solution. For a diffusion the log-growth of a position of size f is $g(f) = f\mu - \frac{1}{2}f^2\sigma^2$ (an arithmetic gain minus a variance penalty):

$$g'(f) = \mu - f\sigma^2 = 0 \;\Rightarrow\; f^\star = \frac{\mu}{\sigma^2} = \frac{0.08}{0.04} = \boxed{2.0}$$

$$g^\star = \frac{\mu^2}{2\sigma^2} = \frac{0.0064}{0.08} = \boxed{8\%} = \tfrac{1}{2}\,(\text{Sharpe})^2$$

Half-Kelly $f = 1$: $g = 0.08 - 0.02 = 6\%$ (75% of the maximum). Twice Kelly $f = 4$: $g = 0.32 - 0.32 = 0$.

⋆ *Analogy:* volatility is a *tax on compounding* whose rate rises with the *square* of your leverage. The optimum is where the marginal tax equals the marginal gain.

B. Fractional Kelly and Estimation Noise

Example 8 — When Your Edge Is Overestimated

You believe $p = 0.6$ on an even-money bet and bet the Kelly fraction $f = 0.2$, but the true probability is $p = 0.55$. Compute the true growth rate at $f = 0.05, 0.1, 0.15, 0.2, 0.25$.
Solution. True Kelly is $f^\star = 2(0.55) - 1 = 0.1$. With $G(f) = 0.55\ln(1+f) + 0.45\ln(1-f)$:

f	.05	.10	.15	.20	.25
G	.00375	.00501	.00374	−.00014	−.00673

Betting 0.2 (double the true Kelly) gives about *zero* growth, and 0.25 loses money despite a real edge. Betting 0.1, i.e. *half* the believed Kelly, lands on the true optimum.
▶ *Speed trick:* the loss from overbetting by Δ is the same as from underbetting by Δ (note 0.05 and 0.15 both give 0.0037), but overbetting by a larger factor falls off the cliff. Errors in p are amplified: a 5-point error in p changes $f^\star$ by 100%.
⋆ *Analogy:* navigating with a GPS that may be off by a few metres: slow down near the edge.

Example 9 — The Fractional-Kelly Trade-Off: $c(2-c)$

Show $G(cf^\star) \approx G^\star\, c(2-c)$. For $p = 0.6$ (even odds) find the fraction c that keeps 90% of the growth.
Solution. Expand G to second order around its maximum, where $G'(f^\star) = 0$:

$$G(f) \approx G^\star - \tfrac{1}{2}|G''|(f - f^\star)^2$$

With $f = cf^\star$ and $G^\star \approx \tfrac{1}{2}|G''|f^{\star 2}$ (the same curvature gives $G(0) = 0$):

$$G(cf^\star) \approx G^\star\left[1 - (c-1)^2\right] = G^\star\, c(2-c)$$

c	0.25	0.5	1	1.5	2
approx	.4375	.75	1	.75	0
exact	.435	.747	1	.733	−.12

For 90%: $c(2-c) = 0.9 \Rightarrow c = 1 - \sqrt{0.1} = 0.684$ (exact value 0.687). So betting about 0.69× Kelly gives up only 10% of growth while cutting position size by 31%.
⋆ *Analogy:* a *hilltop*: the top is flat, so walking down 30% of the way costs 10% of the height, but the cliff on the far side is steeper than the slope you came up.

Example 10 — Half-Kelly Cuts the Chance of Being Behind

Even-money bets with $p = 0.6$, 100 bets. For full Kelly ($f = 0.2$) and half-Kelly ($f = 0.1$) find the per-bet mean and standard deviation of $\ln W$ and $\mathbb{P}(W_{100} < 1)$.
Solution. A win adds $u = \ln(1+f)$, a loss adds $\ln(1-f)$; mean G, standard deviation $\sqrt{pq}\,\left|\ln\frac{1+f}{1-f}\right|$:

$$f = 0.2:\ G = 0.02014,\ \text{sd} = 0.4899(0.4055) = 0.1986$$

$$f = 0.1:\ G = 0.01504,\ \text{sd} = 0.4899(0.2007) = 0.0983$$

Normal approximation $\Phi\big(-\sqrt{n}\,G/\text{sd}\big)$ gives $\Phi(-1.01) = 15.5\%$ and $\Phi(-1.53) = 6.3\%$. Exact binomial: $W_{100} < 1$ needs $k \le 55$ wins at $f = 0.2$ and $k \le 52$ at $f = 0.1$:

$$\mathbb{P}_{\text{full}} = \mathbb{P}(\text{Bin} \le 55) = \boxed{17.9\%}, \qquad \mathbb{P}_{\text{half}} = \mathbb{P}(\text{Bin} \le 52) = \boxed{6.4\%}$$

Half-Kelly keeps 75% of the growth but makes a losing stretch nearly three times less likely.
▶ *Speed trick:* risk scales with f (standard deviation halves) while growth scales like $c(2-c)$. That asymmetry is the entire case for fractional Kelly.

C. Volatility Drag and Leverage

Example 11 — +50% Then −50%

An asset alternately gains 50% and loses 50%. Find the arithmetic mean return, the compound growth factor per period, and compare the drag with $\sigma^2/2$.
Solution. Arithmetic mean: $\frac{1}{2}(+0.5 - 0.5) = 0$. But over two periods:

$$1.5 \times 0.5 = 0.75, \qquad \text{geometric factor} = \sqrt{0.75} = 0.866$$

$$\text{compound return} = \boxed{-13.4\%}\ \text{per period}, \qquad \ln 0.866 = -0.1438$$

Variance of the return is $\sigma^2 = 0.25$ so the approximation drag $\approx \frac{1}{2}\sigma^2 = 0.125$ is close to 0.1438 (the approximation is accurate for small returns).
Recovery asymmetry: after a −50% loss you need +100% to get back to even; after −20%, +25%.
⋆ *Analogy:* climb up half the ladder and then slip down half of what you now have: you are lower than where you began. Averages of returns ignore the order and the compounding.

Example 12 — $g = \mu - \frac{1}{2}\sigma^2$ and Leveraged Decay

An asset has arithmetic mean $\mu = 10\%$ per year and volatility $\sigma = 40\%$. Find (a) the compound growth rate; (b) the breakeven volatility; (c) the optimal leverage and growth when μ is excess return; (d) growth at 2× and 3× leverage.
Solution. (a) Volatility drag is $\frac{1}{2}\sigma^2 = 0.08$:

$$g = \mu - \tfrac{1}{2}\sigma^2 = 0.10 - 0.08 = \boxed{2\%}$$

(b) $g = 0 \iff \sigma = \sqrt{2\mu} = \sqrt{0.2} = 44.7\%$.
(c) Leverage ℓ gives $g(\ell) = \ell\mu - \frac{1}{2}\ell^2\sigma^2$:

$$\ell^\star = \frac{\mu}{\sigma^2} = \frac{0.10}{0.16} = 0.625, \qquad g^\star = \frac{\mu^2}{2\sigma^2} = 0.03125$$

(d) $\ell = 2$: $0.20 - 0.32 = \boxed{-12\%}$; $\ell = 3$: $0.30 - 0.72 = -42\%$. Growth is zero at $\ell = 2\ell^\star = 1.25$: leveraging beyond 1.25× *loses* money even though the asset has a positive mean.
⋆ *Analogy:* a leveraged ETF is a *treadmill*: you may run faster (higher mean) but if the ground shakes too much (variance) you go backwards.

Example 13 — When Kelly Says "Borrow": +20% or −10%

Each period returns +20% or −10% with equal probability. Compute the arithmetic and geometric means, the Kelly fraction, and the growth at several fractions.
Solution. Arithmetic mean: $\frac{1}{2}(0.2 - 0.1) = 5\%$. Geometric: $\sqrt{1.2 \times 0.9} - 1 = \sqrt{1.08} - 1 = 3.92\%$. Kelly with $a = 0.2$, $d = 0.1$:

$$f^\star = \frac{p}{d} - \frac{q}{a} = \frac{0.5}{0.1} - \frac{0.5}{0.2} = 5 - 2.5 = \boxed{2.5}$$

Invest 250% of capital (borrow 150%). Growth $G(f) = \frac{1}{2}\ln(1 + 0.2f) + \frac{1}{2}\ln(1 - 0.1f)$:

f	1	2.5	3	4	5
G	.0385	.0589	.0567	.0385	0

At $f = 2f^\star = 5$, growth is exactly 0: $\frac{1}{2}\ln 2 + \frac{1}{2}\ln\frac{1}{2} = 0$.
▶ *Speed trick:* G is symmetric about $f^\star$ here, giving $G(1) = G(4)$.
Caution: the answer assumes the *worst* loss is exactly 10% per period. A single −40% period at $f = 2.5$ would wipe out all capital (hence real traders cap leverage far below Kelly).

D. Risk of Ruin and Bankroll Management

Example 14 — Gambler's Ruin With a Target

You bet one unit per round on an even-money bet won with probability $p = 0.55$. Starting with 10 units, find the probability of reaching 20 before 0. Compare with a fair game.
Solution. Let $h(k)$ be the probability of reaching N from k. Conditioning on the first bet:

$$h(k) = p\,h(k+1) + q\,h(k-1), \qquad h(0) = 0,\ h(N) = 1$$

The solutions are $A + Br^k$ with $r = q/p$. Imposing the boundaries:

$$h(k) = \frac{1 - r^k}{1 - r^N}, \qquad r = \frac{0.45}{0.55} = \tfrac{9}{11} = 0.818$$

$$r^{10} = 0.1344,\ r^{20} = 0.0181 \ \Rightarrow\ h(10) = \frac{0.8656}{0.9819} = \boxed{0.8815}$$

A fair game gives $\frac{10}{20} = 0.5$. A 5-point edge lifts the doubling probability from 50% to 88%.
⋆ *Analogy:* a drunk stumbling between two walls with a slight tailwind: over many steps the tailwind dominates.

Example 15 — How Many Units Do You Need? (Ruin With No Target)

With a positive edge and unit bets, the probability of *ever* being ruined from k units is r^k with $r = q/p$. Find $\mathbb{P}(\text{ruin})$ for $p = 0.6$, $k = 5$, and the smallest bankroll (in units) giving ruin $\le 1\%$ for $p = 0.6$ and $p = 0.55$.

Solution. (a) $p = 0.6$, $r = \frac{2}{3}$: $\mathbb{P} = (\frac{2}{3})^5 = \boxed{0.132}$.

(b) Need $r^k \le 0.01$, i.e. $k \ge \dfrac{\ln 0.01}{\ln r}$:

$$p = 0.6: \quad \frac{-4.605}{-0.405} = 11.36 \;\Rightarrow\; k = 12 \text{ units} \quad ((\tfrac{2}{3})^{12} = 0.77\%)$$

$$p = 0.55: \quad \frac{-4.605}{-0.2007} = 22.9 \;\Rightarrow\; k = 23 \text{ units} \quad (0.99\%)$$

Halving the edge from 20 to 10 points doubles the bankroll required.

▶ *Speed trick:* ruin probability decays *geometrically* in the bankroll, so a few extra units buy a lot of safety.

⋆ *Analogy:* a moat: each extra unit of width multiplies the chance the invader never crosses by the same factor r.

Example 16 — The Fair Game: Hitting Probability and Duration

A fair ± 1 walk starts at 10 and stops at 0 or 30. Find $\mathbb{P}(\text{hit } 30)$ and the expected number of steps.

Solution. S_n is a martingale, so $\mathbb{E}[S_T] = 10$:

$$30\,\mathbb{P}(\text{hit } 30) + 0 = 10 \;\Rightarrow\; \mathbb{P} = \boxed{\tfrac{1}{3}}$$

Also $S_n^2 - n$ is a martingale:

$$\mathbb{E}[S_T^2] - \mathbb{E}[T] = 10^2, \qquad \mathbb{E}[S_T^2] = 30^2 \cdot \tfrac{1}{3} = 300$$

$$\mathbb{E}[T] = 300 - 100 = \boxed{200} = k(N-k) = 10 \cdot 20$$

(a 10^5-run simulation gives 0.335 and 199.4).

⋆ *Analogy:* the walker needs a time proportional to the *product* of distances to the two walls: moving the walls twice as far apart makes the walk four times longer.

Example 17 — Drawdown: "Will I Ever Lose Half?"

For a bettor using $c\times$Kelly, the probability of ever falling to a fraction α of the starting wealth is approximately $\alpha^{2/c-1}$. Evaluate for $c = 1, \frac{1}{2}, \frac{1}{4}$ with $\alpha = \frac{1}{2}$, and test it on the $p = 0.6$ even-money bet.

Solution. Derivation (diffusion limit): $\ln W$ has drift $\mu_L = f\mu - \frac{1}{2}f^2\sigma^2$ and variance rate $f^2\sigma^2$. The probability that a drifted Brownian motion ever falls by $a = \ln\frac{1}{\alpha}$ is $e^{-2\mu_L a/(f\sigma)^2}$. With $f = c\mu/\sigma^2$, the exponent simplifies to $\left(\frac{2}{c} - 1\right)a$, so $\mathbb{P} = \alpha^{2/c-1}$.

$$c = 1: \ \tfrac{1}{2} \qquad c = \tfrac{1}{2}: \ (\tfrac{1}{2})^3 = \tfrac{1}{8} = 12.5\% \qquad c = \tfrac{1}{4}: \ (\tfrac{1}{2})^7 = 0.78\%$$

Full Kelly has a 50% chance of eventually losing half the bankroll!

Check with real bets ($p = 0.6$): the exact Cramér exponent θ solving $p(1+f)^{-\theta} + q(1-f)^{-\theta} = 1$ is 1 (full), 3.03 (half) and 7.09 (quarter), giving bounds 0.5, 0.122, 0.0073. Simulation of the actual bets gives 0.454, 0.103, 0.006 (slightly below the bound because of finite step sizes).

▶ *Speed trick:* each time you halve the Kelly fraction, the exponent $\frac{2}{c} - 1$ goes $1 \to 3 \to 7$: risk of deep drawdown collapses faster than growth is lost.

⋆ *Analogy:* full Kelly is a mountain pass with a fifty-fifty chance of seeing the valley floor on the way; half-Kelly adds a guard rail.

Example 18 — Streak Risk With Fixed-Fraction Sizing

A trader risks 2% of the current bankroll per trade and wins with probability 0.55 (loss $= -2\%$). How many consecutive losses create a 20% drawdown? What is the probability such a streak occurs in 1000 trades? And with 1% risk?

Solution. After m losses wealth is 0.98^m:

$$0.98^{11} = 0.8007 > 0.8, \qquad 0.98^{12} = 0.7847 \le 0.8 \;\Rightarrow\; \boxed{12 \text{ losses}}$$

A given streak of 12 has probability $0.45^{12} = 6.9 \times 10^{-5}$. A dynamic program tracking the current run of losses gives the probability of at least one such run:

$$\text{in 250 trades: } 0.9\%, \qquad \text{in 1000 trades: } \boxed{3.7\%}$$

(the quick estimate $1000(0.55)(0.45^{12}) = 3.8\%$ agrees). For comparison, a run of 10 losses in 1000 trades has probability 17%.

At 1% risk: $\ln 0.8/\ln 0.99 = 22.2$, so 23 losses are needed; the probability in 1000 trades is only $\approx 6 \times 10^{-6}$.

⋆ *Analogy:* a hundred-year flood: improbable in any one year, but sized into your plan over a career. Size so that the streak you *will* see does not kill you.

E. Capstones

Example 19 — Two Simultaneous Bets

Each round you can bet on two *independent* even-money bets, each won with probability 0.6. You stake the same fraction f on each. Find the Kelly fraction and the growth rate.

Solution. Both win (0.36): wealth $\times(1+2f)$; one each (0.48): $\times 1$; both lose (0.16): $\times(1-2f)$:

$$G(f) = 0.36\ln(1+2f) + 0.16\ln(1-2f)$$

$$\frac{0.36}{1+2f} = \frac{0.16}{1-2f} \;\Rightarrow\; 0.36 - 0.72f = 0.16 + 0.32f$$

$$f = \frac{0.2}{1.04} = \frac{5}{26} = \boxed{0.1923}$$

Total exposure is 0.385 (not 0.4) and

$$G = 0.36\ln 1.3846 + 0.16\ln 0.6154 = \boxed{0.0395}$$

versus 0.0201 for one bet. Two independent bets nearly double the growth (0.0395 against $2 \times 0.0201 = 0.0403$ if the bets were placed in sequence).

⋆ *Analogy:* two independent engines: you can open each throttle a little less than you would for one engine, but together you travel almost twice as fast.

Example 20 — Time to Double: Small Edges Are Slow

At even odds with win probability $p = \frac{1+\varepsilon}{2}$ the Kelly fraction is $f^\star = \varepsilon$ and $G \approx \varepsilon^2/2$. Find the median number of bets to double the bankroll for $\varepsilon = 0.01$ and $\varepsilon = 0.02$.

Solution. $\ln 2 \approx nG \Rightarrow n \approx \dfrac{2\ln 2}{\varepsilon^2}$:

$$\varepsilon = 0.01 \ (p = 0.505): \ G = 5.0 \times 10^{-5}, \ n \approx \boxed{13{,}860}$$

$$\varepsilon = 0.02 \ (p = 0.51): \ G = 2.0 \times 10^{-4}, \ n \approx \boxed{3{,}470}$$

(exact values from $G(f) = p\ln(1+f) + q\ln(1-f)$: 13,863 and 3,466). Doubling the edge makes the time *four times shorter*.

▶ *Speed trick:* growth rate $\propto \varepsilon^2$, so edge estimation matters quadratically.

⋆ *Analogy:* a small edge is a slow river: it eventually fills the lake, but a stream with twice the flow fills it in a quarter of the time.

Example 21 — Capstone: Sizing a 55% Strategy

A strategy wins 55% of even-money bets. Compare full Kelly with half-Kelly on growth, time to double, risk of being behind, drawdown, and robustness to an estimation error.

Solution. $f^\star = 2(0.55) - 1 = 0.1$. With $u = \ln(1+f)$ and $G = 0.55u + 0.45\ln(1-f)$:

	$f = 0.1$	$f = 0.05$
G	.005008	.003753
sd per bet	.0998	.0498
median bets to double	138	185

Half-Kelly keeps 74.9% of the growth.

Being behind after 200 bets: full $\mathbb{P} = 26.1\%$, half $\mathbb{P} = 14.3\%$ (exact binomial).

Ever losing half the bankroll (Example 17): bound 0.50 and 0.125; simulation of the actual bets gives 47.6% and 11.4%.

Robustness: suppose the true win probability is 0.52 (edge overestimated). Then

$$G(0.1) = -0.00101 \text{ (loses)}, \qquad G(0.05) = +0.00075 \text{ (still wins)}$$

Decision: half-Kelly gives up 25% of the best-case growth but cuts the drawdown risk by about $4\times$ and survives a 3-point error in p.

⋆ *Full-circle intuition:* Kelly is the *speed limit* for compounding. You never want to drive above it, and when your speedometer is uncertain you drive well below it.